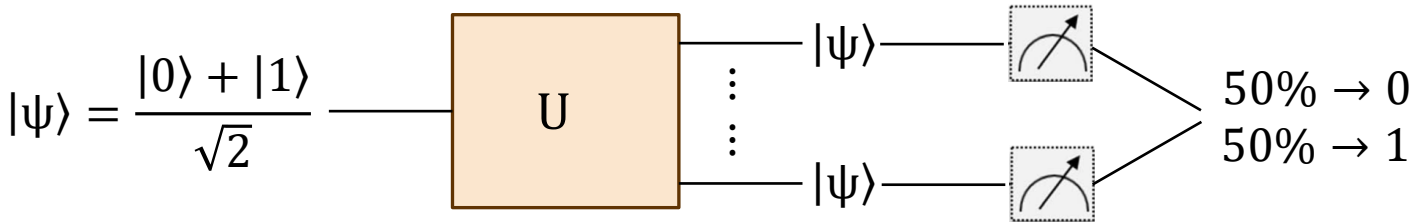


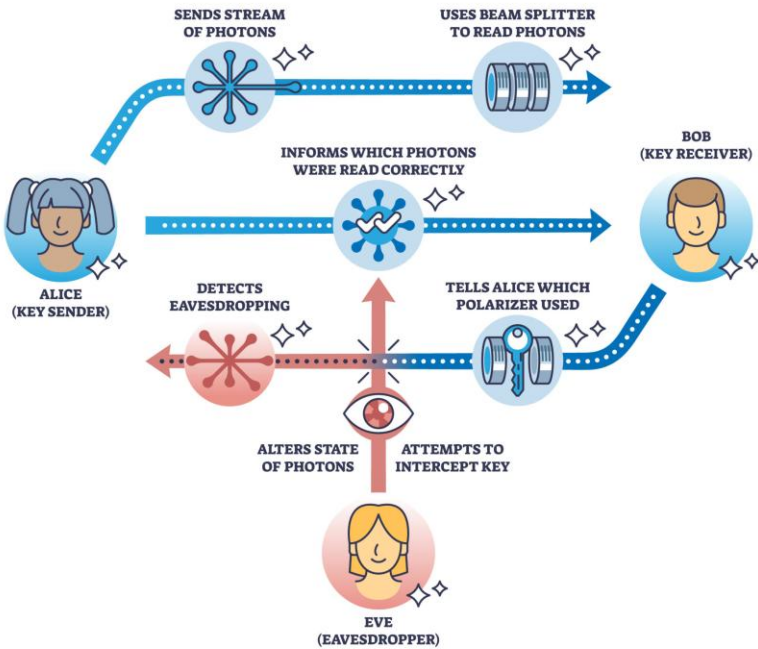
Introduction to Quantum Computing (L-5)

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Recap- No Cloning of an Arbitrary Quantum Bit



PROCESS OF QUANTUM KEY DISTRIBUTION (QKD)



If cloning were possible:

- Clone many times
- Measure all copies
- Reconstruct α and β exactly

That would let us:

- Extract full quantum information
- Violate quantum uncertainty
- Break quantum mechanics

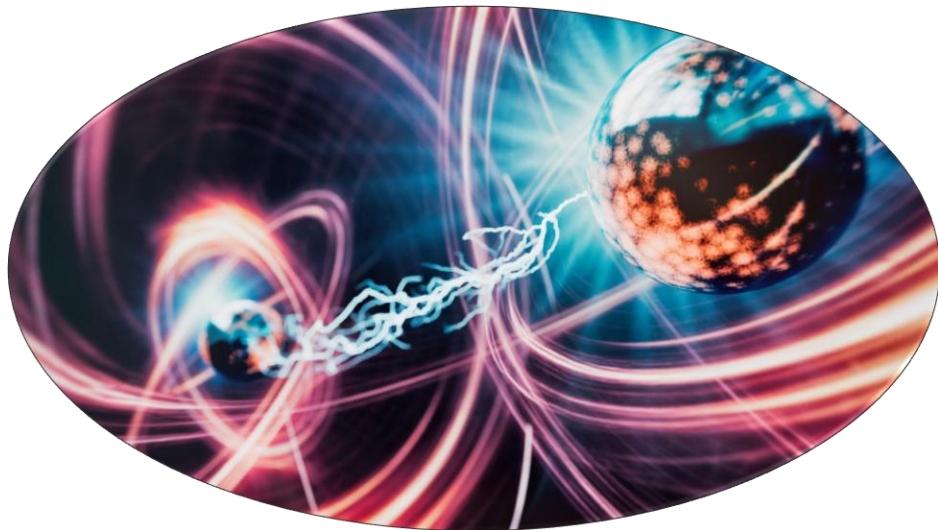
Nature prevents this by **forbidding cloning**

Quantum Cryptography (BB84): Because qubits cannot be copied, any eavesdropping attempt necessarily disturbs the state and reveals itself, making security guaranteed by physics rather than computational assumptions.

Quantum Teleportation: The no-cloning principle ensures that the quantum state is destroyed at the sender and recreated at the receiver, preventing any illegal duplication of quantum information.

Quantum Error Correction: Since qubits cannot be backed up by copying, error correction relies on entanglement and carefully distributed redundancy, making it fundamentally different from classical error-correction schemes.

Recap- Entanglement



A quantum state of two qubits, $|\psi\rangle = \alpha|00\rangle + \beta|11\rangle$ is said to be entangled if it cannot be expressed as a tensor product of the quantum states of the individual qubits. For example, $|\psi\rangle = \sqrt{0.9}|00\rangle \pm \sqrt{0.1}|11\rangle$ is impossible to express as a tensor product of two individual qubit states.

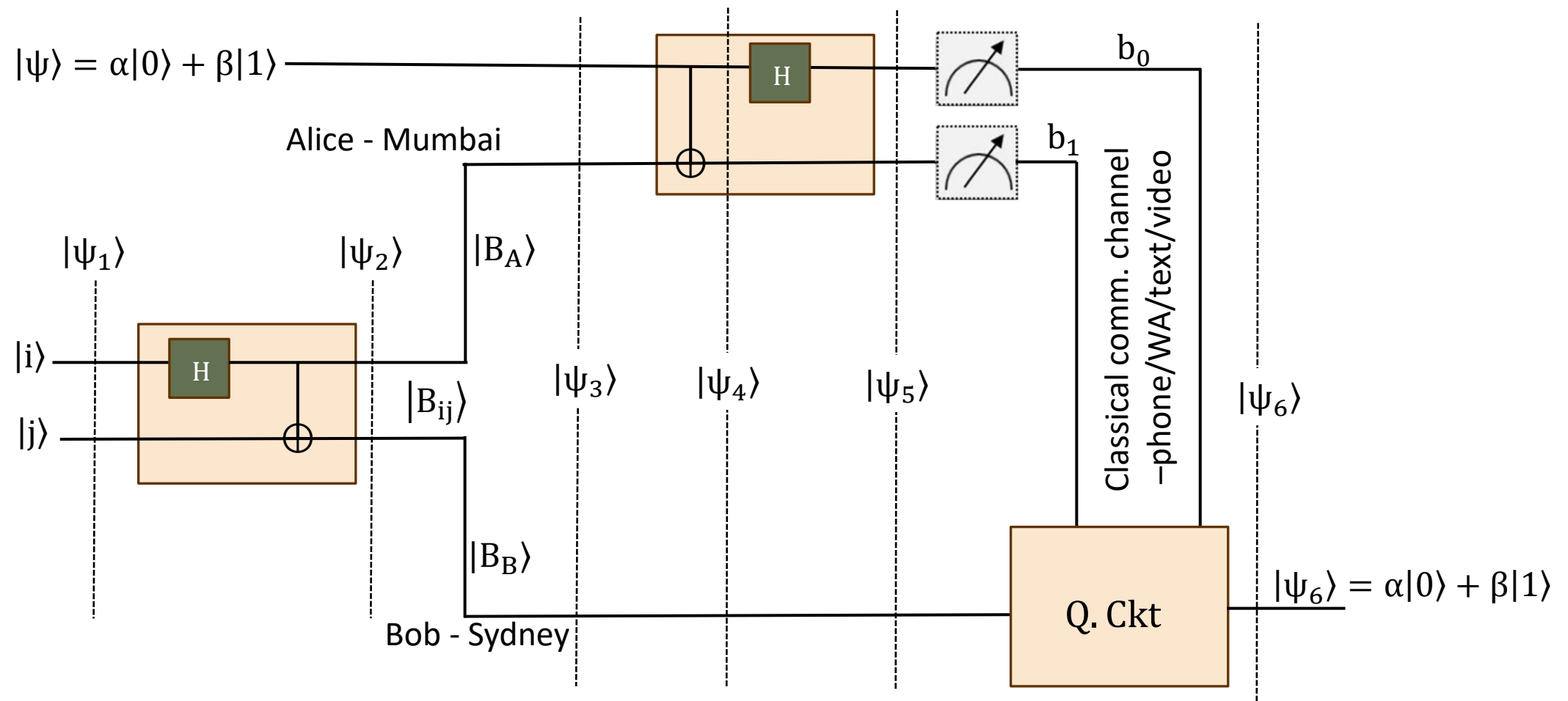
However, there are a special class of entangled state which are maximally entangled, or rather, the individual qubits participating in the entanglement share the strongest correlation. This happens when $\alpha = \beta = 1/\sqrt{2}$. For any non-equal (α, β) the correlation is only partial.

Such maximally entangled states are called Bell states. Sometimes people also call them the Einstein–Podolsky–Rosen or EPR states.

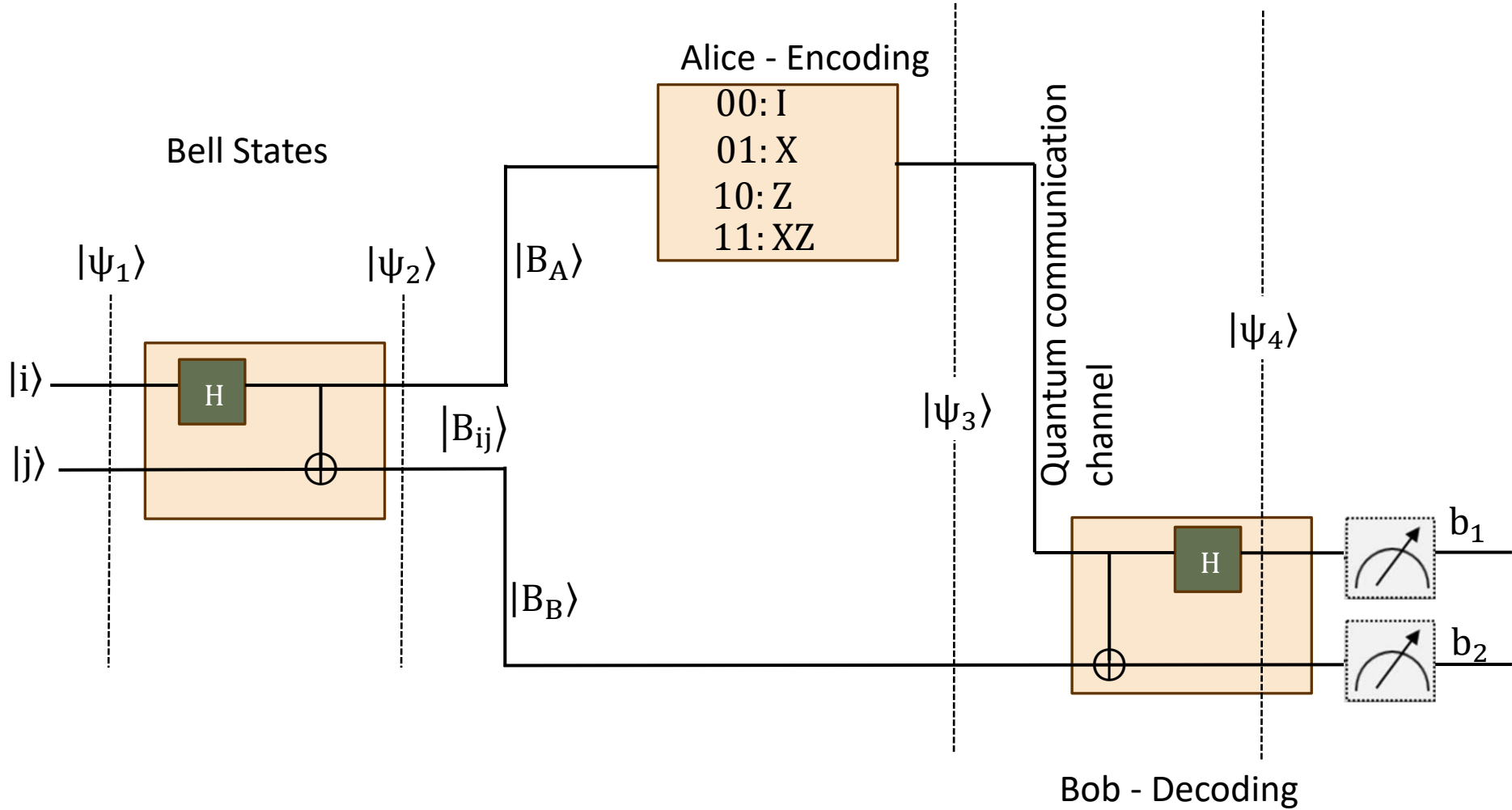
$$\left. \begin{aligned} |\phi^+\rangle &= |B_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\ |\phi^-\rangle &= |B_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ |\psi^+\rangle &= |B_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \\ |\psi^-\rangle &= |B_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \end{aligned} \right\} \text{Bell states}$$

- Smallest system where quantum correlations beat classical ones
- First place where “quantum advantage” appears
- Minimal system that violates Bell inequalities - if a device can’t create Bell states reliably, it’s not truly quantum

Recap – Quantum Teleportation



Recap – Superdense Coding



- For 00: $|\psi_3\rangle = I|\psi_2\rangle$
- For 01: $|\psi_3\rangle = X|\psi_2\rangle$
- For 10: $|\psi_3\rangle = Z|\psi_2\rangle$
- For 11: $|\psi_3\rangle = XZ|\psi_2\rangle$

Classical Circuits and Boolean Functions

Any Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$ can be implemented using a classical circuit containing classical gates like NOT, AND, OR, or simply any universal gate like NAND or NOR

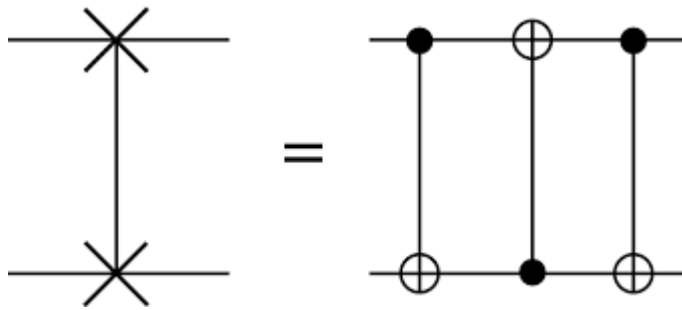
If $f(00) = 1$, $f(11) = 1$, and 0 otherwise. This can be implemented using the NOT, AND, OR gate as
$$f(x_1, x_2) = \overline{x_1} \cdot \overline{x_2} + (x_1 \cdot x_2)$$

If $f(01) = 1$, $f(10) = 1$, and 0 otherwise. This can be implemented using the NOT, AND, OR gate as
$$f(x_1, x_2) = \overline{x_1} \cdot x_2 + x_1 \cdot \overline{x_2}$$

If $f(00) = 1$, $f(01) = 1$, and 0 otherwise. This can be implemented using the NOT, AND, OR gate as
$$f(x_1, x_2) = \overline{x_1} \cdot \overline{x_2} + \overline{x_1} \cdot x_2$$

We cannot directly convert a classical circuit block-by-block into a quantum circuit because classical gates are generally irreversible, while quantum operations must be reversible (unitary) and information-preserving.

Quantum SWAP Gates



- The SWAP gate is a basic two-qubit quantum logic gate whose action is to exchange (swap) the quantum states of two qubits: $|a, b\rangle \xrightarrow{\text{SWAP}} |b, a\rangle$
- Intuitively, if two qubits hold states $|\psi_1\rangle$ and $|\psi_2\rangle$, after SWAP they are simply interchanged
- This gate is used widely:
 - To move quantum information between qubits, especially in architectures with limited qubit connectivity.
 - In quantum communication and algorithms (e.g., the SWAP test).

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{SWAP}|01\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle$$

$$|00\rangle \xrightarrow{\text{SWAP}} |00\rangle$$

$$|01\rangle \xrightarrow{\text{SWAP}} |10\rangle$$

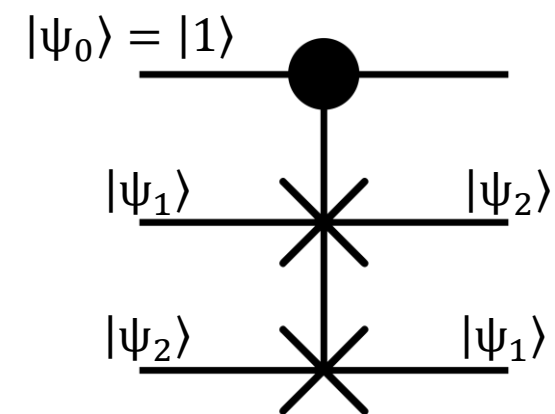
$$|10\rangle \xrightarrow{\text{SWAP}} |01\rangle$$

$$|11\rangle \xrightarrow{\text{SWAP}} |11\rangle$$

$$(\text{SWAP})^{-1} = (\text{SWAP})^\dagger$$

This matrix is unitary (as all quantum logic gates must be) and its own inverse

Controlled-SWAP (CSWAP) Gates



The controlled-SWAP (CSWAP) gate swaps two target qubits only if a control qubit is in state $|1\rangle$. CSWAP swaps qubits $|\psi_1\rangle$ and $|\psi_2\rangle$ iff the control qubit $|\psi_0\rangle = |1\rangle$

$$\text{CSWAP} = \begin{bmatrix} I_4 & 0_4 \\ 0_4 & \text{SWAP} \end{bmatrix}_{8 \times 8}$$

$$\text{where SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and } 0_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(\text{CSWAP})^{-1} = (\text{CSWAP})^\dagger$$

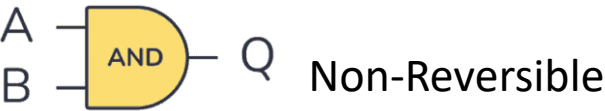
$|c, a, b\rangle \xrightarrow{\text{CSWAP}} |0, a, b\rangle$ Control qubit = $|0\rangle \rightarrow$ do nothing
 $|1, b, a\rangle$ Control qubit = $|1\rangle \rightarrow$ swap target qubits

- Quantum data comparison: CSWAP is the core gate behind the SWAP test, used to compare quantum states.
- Reversible classical computing: Fredkin gate is universal for reversible classical logic.
- Quantum algorithms: Used in State overlap estimation, Quantum machine learning, Quantum fingerprinting

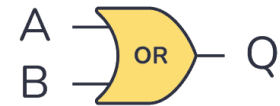
C-SWAP Gates as Reversible Analogues of Classical Gates



A	Q
0	1
1	0

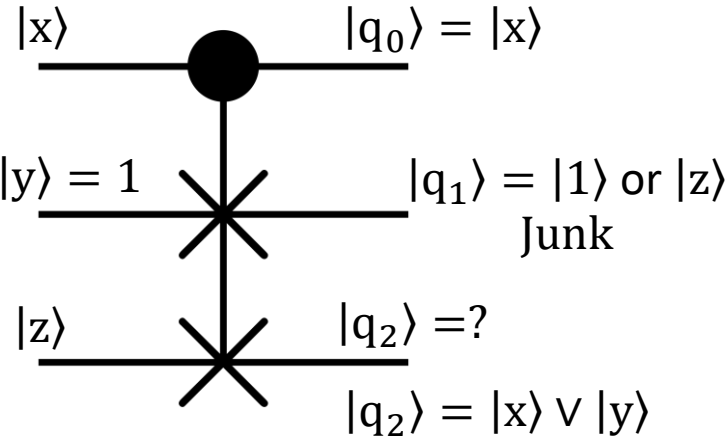
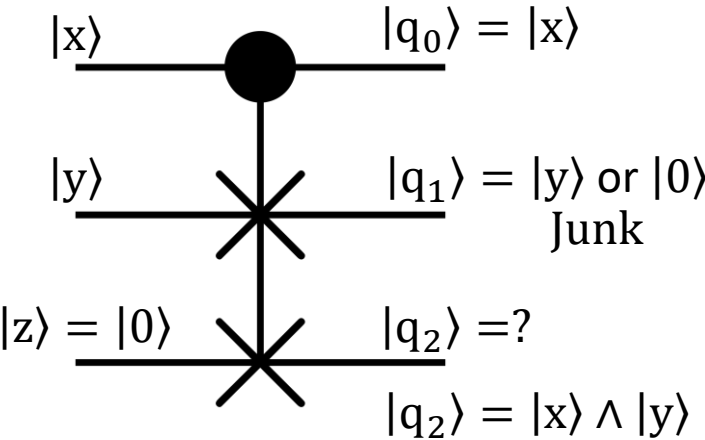


A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

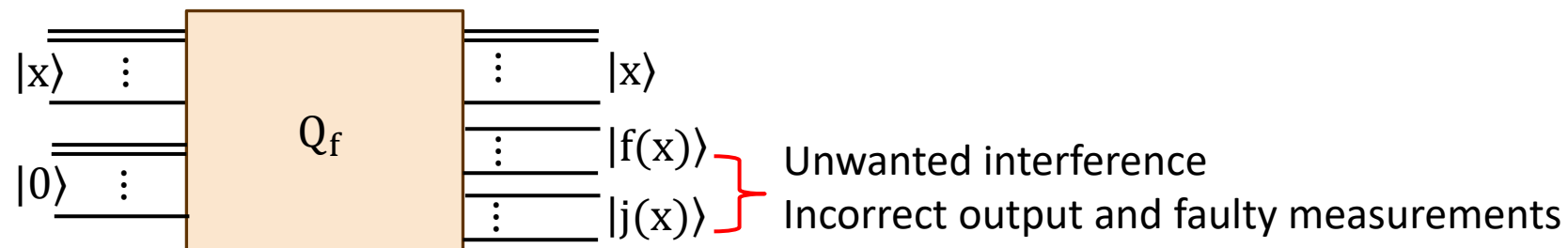
Non-Reversible



$ x\rangle$	$ y\rangle$	$ z\rangle$	$ q_0\rangle$	$ q_1\rangle$	$ q_2\rangle$
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$
$ 1\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$

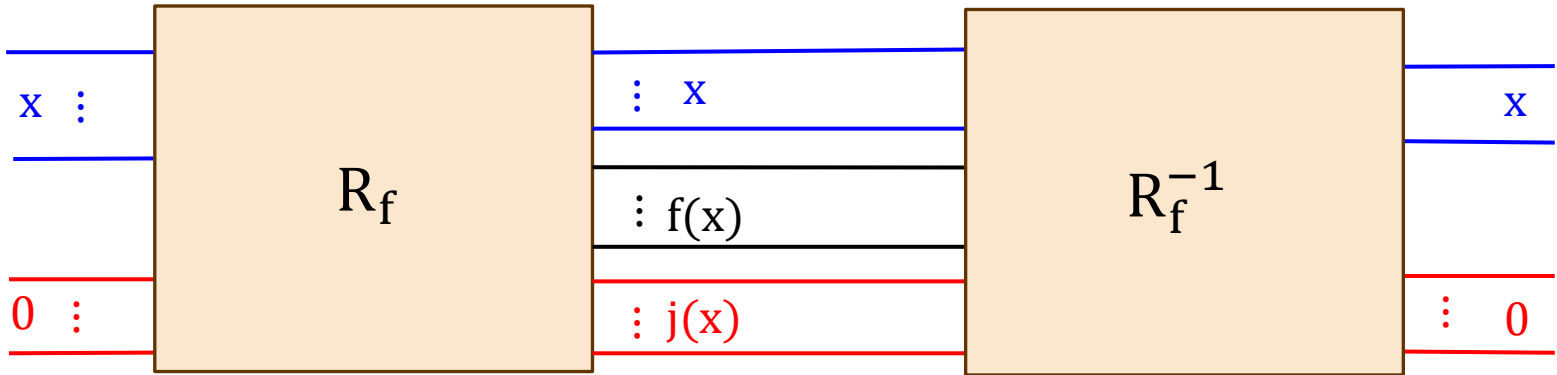
$ x\rangle$	$ y\rangle$	$ z\rangle$	$ q_0\rangle$	$ q_1\rangle$	$ q_2\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 1\rangle$

Reversible Logic: From Classical Gates to Oracles

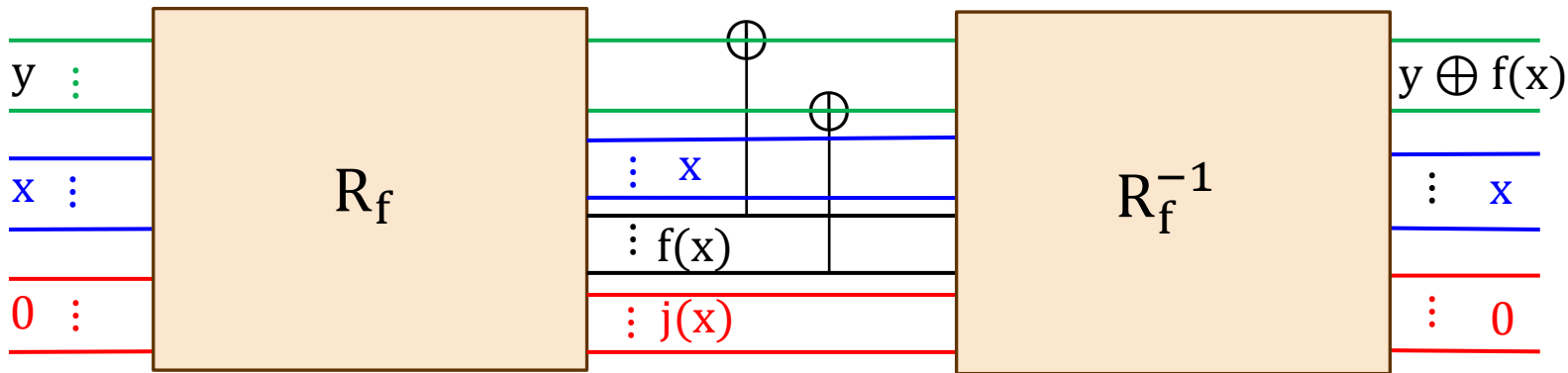


- Using reversible gates like CSWAP (Fredkin) and NOT, we can encode any classical function $f(x)$ inside a reversible circuit Q_f .
- However, to keep the transformation one-to-one, the circuit also produces extra outputs $j(x)$, called junk (or garbage) bits.
- In a quantum setting, these junk bits cannot be ignored. They may entangle or remain in superposition with the useful output $f(x)$.
- This unwanted correlation can create undesired interference, leading to incorrect results when we measure.
- Unlike classical circuits, we cannot simply discard or block $j(x)$; quantum circuits must preserve all information due to reversibility.

Reversible Logic: From Classical Gates to Oracles



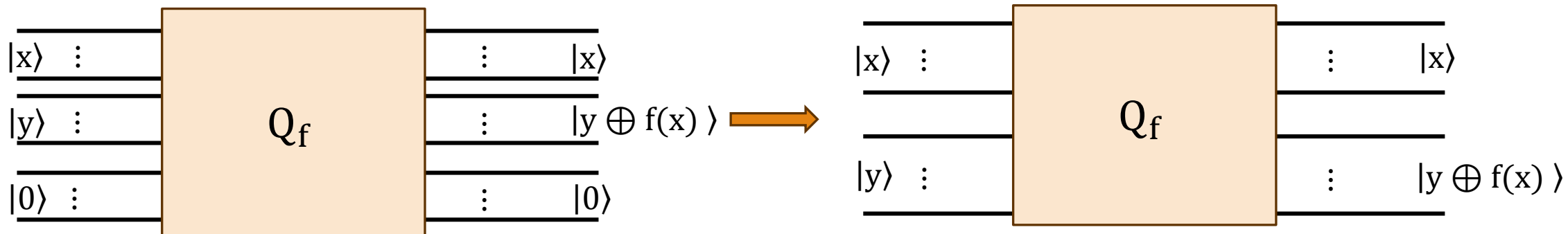
- If we feed the outputs of R_f into its inverse circuit R_f^{-1} , the junk bits $j(x)$ are removed.
- However, reversing the circuit also erases the desired output $f(x)$, because we undo the entire computation.
- So simply applying R_f^{-1} is not enough. We lose both the junk and the function value.
- The key idea is: we must store $f(x)$ somewhere else first, before applying the inverse.



- Before applying R_f^{-1} , we reversibly copy $f(x)$ into an ancilla register using an XOR operation: $y \rightarrow y \oplus f(x)$
- After storing the function value, we run R_f^{-1} to remove the junk $j(x)$ while preserving both x and $y \oplus f(x)$.

Reversal only undoes the operations that were part of that specific circuit. Since the XOR was done after R_f , and not inside it, the inverse does not affect it.

Reversible Logic: From Classical Gates to Oracles



- Classical circuits can erase information, but quantum circuits must be reversible, so we embed any function $f(x)$ inside a larger reversible circuit.
- The reversible circuit computes $f(x)$ while preserving the input x , but it also produces extra junk bits that cannot simply be discarded.
- We copy the useful output using a reversible XOR, then apply the inverse circuit to remove the junk without erasing the function value.
- In practical quantum circuit diagrams, ancilla qubits that start in $|0\rangle$ and return to $|0\rangle$ after computation are often not shown, since they do not affect the final result

Your First Quantum Algorithm: Deutsch Algorithm

Simple but impractical. No application, but a good point to start understanding the basics of quantum algorithms

Problem definition: Let there be a function f that takes a single bit as input and outputs a single bit as well. $f\{0,1\} \rightarrow \{0,1\}$. We need to find whether the function is “constant” or “balanced”.

A function is called constant if it returns the same output (0 or 1) for all inputs. A function is called balanced if it returns 0 for half the inputs and 1 for the other half.

n = 1

Constant

x	f(x)
0	0
1	0

x	f(x)
0	1
1	1

Balanced

x	f(x)
0	1
1	0

x	f(x)
0	0
1	1

		n = 2		
Constant	x	f(x)	x	f(x)
	00	1	00	0
	01	1	01	0
	10	1	10	0
	11	1	11	0

				Balanced			
x	f(x)	x	f(x)	x	f(x)	x	f(x)
00	0	00	1	00	0	00	1
01	0	01	1	01	1	01	0
10	1	10	0	10	0	10	1
11	1	11	0	11	1	11	0

The classical algorithm needs call the oracle function $(2^n/2) + 1$ time to check if the function is constant or balanced $\Rightarrow O(2^n)$. Quantum algorithm needs call the oracle function once owing to the power of superposition $\Rightarrow O(1)$

First Algorithm: Deutsch Algorithm

Constant

Case: 1	
x	f(x)
0	0
1	0

Case: 2

x	f(x)
0	1
1	1

Balanced

x	f(x)
0	1
1	0

Case: 3

x	f(x)
0	0
1	1

Case: 4

x	f(x)
0	1
1	0

$$|\psi_1\rangle = |01\rangle$$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$= \frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2}$$

Next we apply Q_f to $|\psi_2\rangle$
 Recall: $Q_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$

$|\psi_1\rangle$
 $|\psi_2\rangle$

$|\psi_3\rangle$
 $|\psi_4\rangle$

We make a claim that if we get a '0' after measurement, the function is constant, or else if it measures '1', the function is balanced.

$$|\psi_3\rangle = \frac{1}{2} [|0\rangle|0 \oplus f(0)\rangle - |0\rangle|1 \oplus f(0)\rangle + |1\rangle|0 \oplus f(1)\rangle + |1\rangle|1 \oplus f(1)\rangle]$$

Case 1: $f(0) = f(1) = 0$ $|\psi_3\rangle = \frac{1}{2} [|0\rangle|0\rangle - |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle]$

$$= \frac{1}{2} [|0\rangle\{|0\rangle - |1\rangle\} + |1\rangle\{|0\rangle + |1\rangle\}]$$

$$= \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\} \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} = |+\rangle|-\rangle$$

$|\psi_4\rangle = H|+\rangle|-\rangle = |0\rangle|-\rangle$
 Case 2: $f(0) = f(1) = 1$ $|\psi_4\rangle = H(-1)|+\rangle|-\rangle = -|0\rangle|-\rangle$

Measures '0' with 100% probability

First Algorithm: Deutsch Algorithm

n = 1

Case: 1

x	f(x)
0	0
1	0

Case: 2

x	f(x)
0	1
1	1

Constant

Case: 3

x	f(x)
0	1
1	0

Case: 4

x	f(x)
0	0
1	1

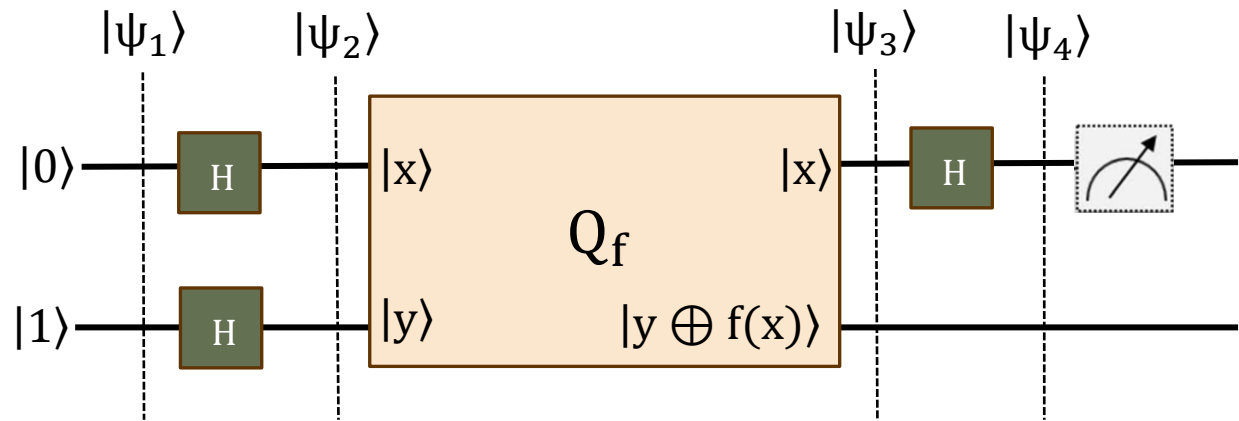
Balanced

$$|\psi_1\rangle = |01\rangle$$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$= \frac{|00\rangle - |01\rangle + |10\rangle - |11\rangle}{2}$$

Next we apply Q_f to $|\psi_2\rangle$
 Recall: $Q_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$



We make a claim that if we get a '0' after measurement, the function is constant, or else if it measures '1', the function is balanced.

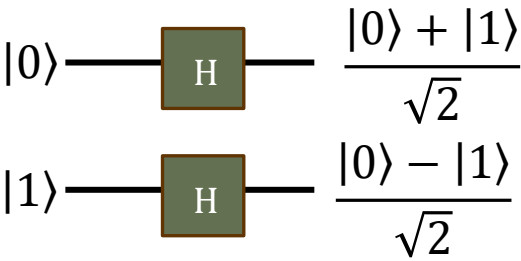
$$|\psi_3\rangle = \frac{1}{2} [|0\rangle|0 \oplus f(0)\rangle - |0\rangle|1 \oplus f(0)\rangle + |1\rangle|0 \oplus f(1)\rangle + |1\rangle|1 \oplus f(1)\rangle]$$

Case 3: $f(0) = 1; f(1) = 0$ $|\psi_3\rangle = \left\{ \frac{-|0\rangle + |1\rangle}{\sqrt{2}} \right\} \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} = -|-\rangle|-\rangle$

Case 4: $f(0) = 0; f(1) = 1$ $|\psi_3\rangle = \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} = |-\rangle|-\rangle$

$$|\psi_4\rangle = H|\psi_3\rangle = \pm|1\rangle|-\rangle$$
 Measures '1' with 100% probability
 Ignoring global phase

Deutsch Jozsa Algorithm



$H|x\rangle_{x \in \{0,1\}} = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}}$
 $(-1)^0 = 1$ and $(-1)^1 = -1$

General form for single qubit

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{z \in \{0,1\}} (-1)^{x \cdot z} |z\rangle$$

Hadamard gate on 2-qubit register



$$H^{\otimes 2} |x^{\otimes 2}\rangle = \sum_{z_1 \in \{0,1\}} \frac{(-1)^{x_1 \cdot z_1} |z_1\rangle}{\sqrt{2}} \otimes \sum_{z_2 \in \{0,1\}} \frac{(-1)^{x_2 \cdot z_2} |z_2\rangle}{\sqrt{2}} = \sum_{z_1, z_2 \in \{0,1\}} \frac{(-1)^{(x_1 \cdot z_1 + x_2 \cdot z_2)} |z_1 z_2\rangle}{\sqrt{2^2}} = \sum_{\substack{z \in \{0,1\}^2 \\ \text{or} \\ z \in \{00,01,10,11\}}} \frac{(-1)^{x \cdot z} |z\rangle}{\sqrt{2^2}}$$

Deutsch Jozsa Algorithm

Hadamard gate on 2-qubit register

$$|x^{\otimes 2}\rangle_{x \in \{0,1\}} \xrightarrow{H^{\otimes 2}} H|x_1\rangle \otimes H|x_2\rangle$$

$$H^{\otimes 2} |x^{\otimes 2}\rangle = \sum_{\substack{z \in \{0,1\}^2 \\ \text{or} \\ z \in \{00,01,10,11\}}} \frac{(-1)^{x \cdot z} |z\rangle}{\sqrt{2^2}}$$

Let $x = \{x_1, x_2\} = 00$:

$$(x_1 \cdot z_1 + x_2 \cdot z_2) = 0 \cdot z_1 + 0 \cdot z_2 = 0 \Rightarrow (-1)^{x \cdot z} = (-1)^0 = 1$$

$$H^{\otimes 2} |x^{\otimes 2}\rangle = \sum_{z \in \{0,1\}^2} \frac{|z\rangle}{\sqrt{2^2}} = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2^2}}$$

Let $x = \{x_1, x_2\} = 10$:

$$\text{When } z = \{z_1, z_2\} = 00 \rightarrow x \cdot z = 0 \rightarrow (-1)^{x \cdot z} = +1$$

$$\text{When } z = \{z_1, z_2\} = 01 \rightarrow x \cdot z = 0 \rightarrow (-1)^{x \cdot z} = +1$$

$$\text{When } z = \{z_1, z_2\} = 10 \rightarrow x \cdot z = 1 \rightarrow (-1)^{x \cdot z} = -1$$

$$\text{When } z = \{z_1, z_2\} = 11 \rightarrow x \cdot z = 1 \rightarrow (-1)^{x \cdot z} = -1$$

$$\begin{aligned} H^{\otimes 2} |x^{\otimes 2}\rangle &= \sum_{z \in \{0,1\}^2} \frac{(-1)^{x \cdot z} |z\rangle}{\sqrt{2^2}} = \frac{|00\rangle + |01\rangle - |10\rangle - |11\rangle}{\sqrt{2^2}} \\ &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \end{aligned}$$

Deutsch Jozsa Algorithm

Hadamard gate on n-qubit register

$$|x^{\otimes n}\rangle_{x \in \{0,1\}} \xrightarrow{H^{\otimes n}} H|x_1\rangle \otimes H|x_2\rangle \dots \otimes H|x_n\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle$$

Deutsch’s Algorithm: $f\{0,1\} \rightarrow \{0,1\}$

Deutsch-Jozsa Algorithm: $f\{0,1\}^n \rightarrow \{0,1\}$

Constant

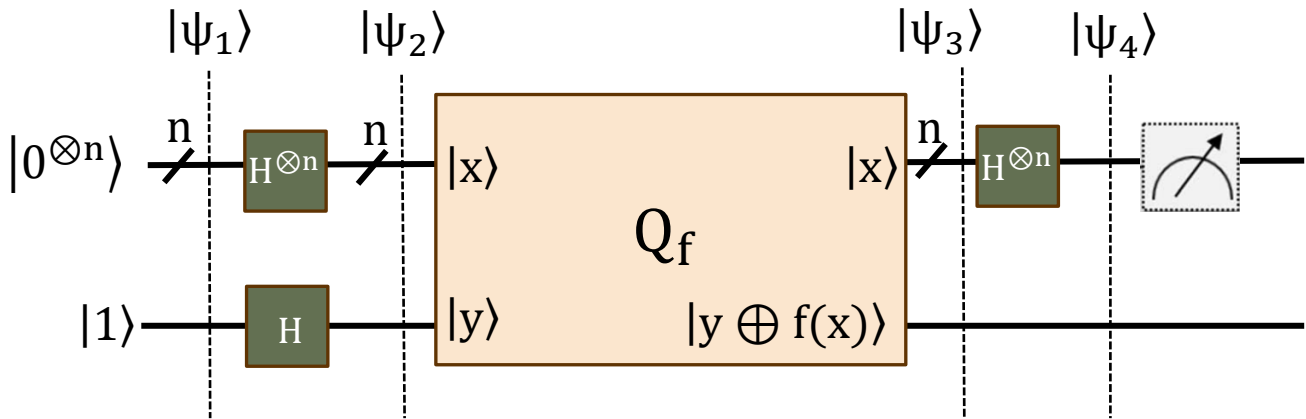
x	$f(x)$	x	$f(x)$
00...00	0	00...00	1
00...01	0	00...01	1
00...10	0	00...10	1
00...11	0	00...11	1
...	...	...	...
11...11	0	11...11	1

For a Balanced function

$f(x) = 0$ for half the inputs and $f(x) = 1$ for the other half of the inputs.

No. of x ’s where $f(x) = 0$ is the same as the no. of x ’s where $f(x) = 1$

Deutsch Jozsa Algorithm



$$|\psi_1\rangle = |0^{\otimes n}\rangle|1\rangle$$

$$|\psi_2\rangle = H^{\otimes n}|0^{\otimes n}\rangle H|1\rangle$$

$$H^{\otimes n}|\xi^{\otimes n}\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{\xi \cdot z} |z\rangle$$

$$H^{\otimes n}|0^{\otimes n}\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Next we apply Q_f to $|\psi_2\rangle$

Recall: $Q_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle \frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}}$$

$$0 \oplus f(x) = f(x) \text{ and } 1 \oplus f(x) = \overline{f(x)}$$

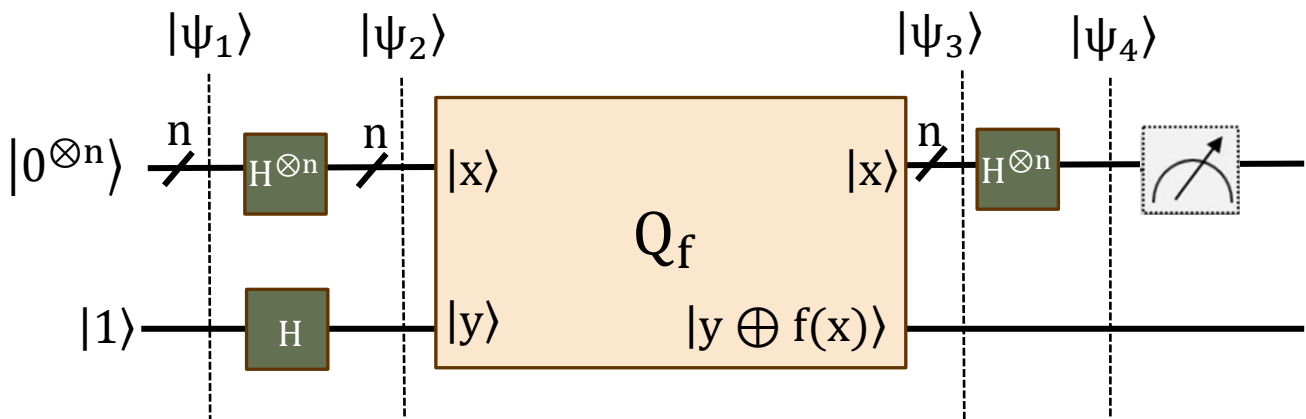
$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle \frac{f(x) - \overline{f(x)}}{\sqrt{2}}$$

if $f(x) = 0$ then $\overline{f(x)} = 1$, the 2nd register reads $\frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$

if $f(x) = 1$ then $\overline{f(x)} = 0$, the 2nd register reads $\frac{-|0\rangle + |1\rangle}{\sqrt{2}} = -|-\rangle$

$$\frac{f(x) - \overline{f(x)}}{\sqrt{2}} = (-1)^{f(x)}|-\rangle$$

Deutsch-Jozsa Algorithm



$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle (-1)^{f(x)} |-\rangle$$

$$|\psi_4\rangle = H^{\otimes n} \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle (-1)^{f(x)} \otimes |-\rangle$$

$$= \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{f(x)} H^{\otimes n} |z\rangle \otimes |-\rangle$$

Next, we shall measure only the first register. So, lets forget about the $|-\rangle$ in the 2nd register.

Let us check the probability of measuring $|u\rangle = |0^{\otimes n}\rangle$ when $f(x)$ is constant. The algorithm claims that this probability will be 100%.

$$|\psi_4\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{u \in \{0,1\}^n} (-1)^{z \cdot u} |u\rangle$$
$$= \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \sum_{u \in \{0,1\}^n} (-1)^{f(x) + z \cdot u} |u\rangle$$

Since $|u\rangle = |0^{\otimes n}\rangle$, then $z \cdot u = 0$

For a constant function, when $f(x) = 0$ we have $(-1)^{f(x)} = (-1)^0 = 1$

$$|\psi_4\rangle_{|u\rangle=|0^{\otimes n}\rangle} = \frac{1 + 1 + 1 \dots \dots + 1}{2^n} |0^{\otimes n}\rangle$$

With ‘n’ bit input string, there are 2^n possible input combinations.

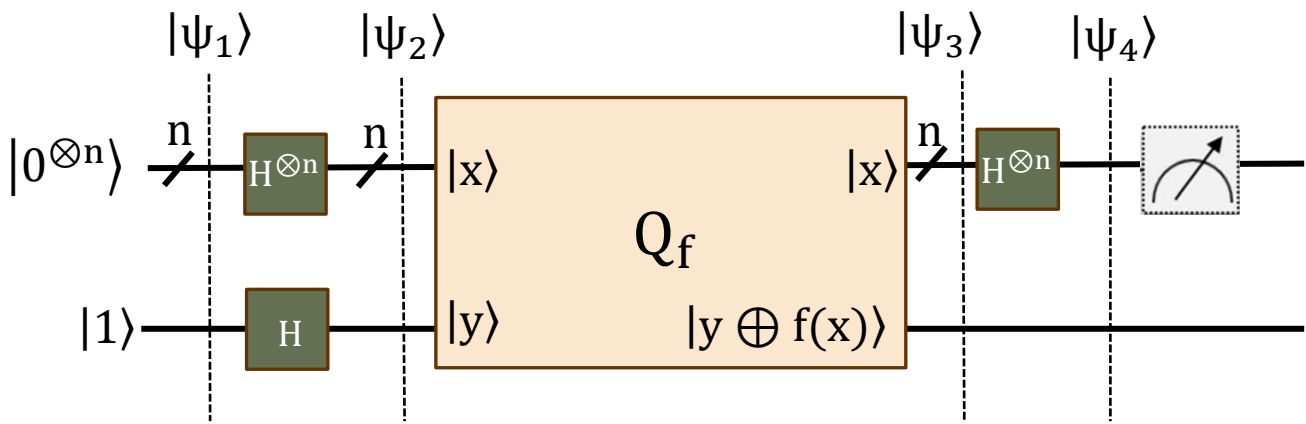
$$|\psi_4\rangle_{|u\rangle=|0^{\otimes n}\rangle} = \frac{2^n}{2^n} |0^{\otimes n}\rangle = |0^{\otimes n}\rangle$$

For a constant function, when $f(x) = 1$ we have $(-1)^{f(x)} = -1$

$$|\psi_4\rangle_{|u\rangle=|0^{\otimes n}\rangle} = \frac{-1 - 1 - 1 \dots \dots - 1}{2^n} |0^{\otimes n}\rangle = \frac{-2^n}{2^n} |0^{\otimes n}\rangle = -|0^{\otimes n}\rangle$$

In either case, the probability of measuring ‘0’ is the mod square of $\pm |0^{\otimes n}\rangle$, which is 100% when $f(x)$ is constant.

Deutsch-Jozsa Algorithm



Let us check the probability of measuring $|u\rangle = |0^{\otimes n}\rangle$ when $f(x)$ is balanced. The algorithm claims that this probability will be 0%.

Since we are trying to estimate the probability that we shall get all 0's at the output all the time, we consider $|u\rangle = |0^{\otimes n}\rangle$, then $z \cdot u = 0$

$$\begin{aligned} |\psi_4\rangle &= \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{u \in \{0,1\}^n} (-1)^{z \cdot u} |u\rangle \\ &= \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \sum_{u \in \{0,1\}^n} (-1)^{f(x) + z \cdot u} |u\rangle \end{aligned}$$

For a balanced function, when $f(x) = 0$ for one half of the inputs and $f(x) = 1$ for the other half. This means, $(-1)^{f(x)} = 1$ for one half of the inputs and $(-1)^{f(x)} = -1$ for the other half.

$$|\psi_4\rangle_{|u\rangle=|0^{\otimes n}\rangle} = \frac{(1 + 1 \dots + 1)_{2^{n/2}} + (-1 - 1 \dots - 1)_{2^{n/2}}}{2^n} |0^{\otimes n}\rangle = 0 |0^{\otimes n}\rangle$$

The probability of measuring '0' which is the mod square of $0 |0^{\otimes n}\rangle$, which is 0% when $f(x)$ is balanced.

So, we will always measure 0 at the output when $f(x)$ is constant and we will never measure 0 at the output when $f(x)$ is balanced

Qiskit – Prepare an Arbitrary Qubit

In this exercise, you will:

1. Start from the computational basis state $|0\rangle$
2. Create an arbitrary single-qubit state using rotations
3. Print the state vector at input and output
4. Plot the Bloch sphere at the input and output
5. Verify that your final state matches the theoretical formula

```
qc = QuantumCircuit(1)
qc.ry(theta, 0)
qc.rz(phi, 0)
```

Do not choose a θ and ϕ that is a multiple of π or $\pi/2$

Example, you may choose something like:

```
theta = np.pi / 3
phi = np.pi / 2
```

Qiskit – CSWAP

In this exercise, you will:

1. Create three qubits
2. Prepare two target qubits in the states $|0\rangle$ and $|1\rangle$
3. Apply a Controlled-SWAP (CSWAP) gate
4. First keep the control qubit in $|0\rangle$
5. Then prepare the control qubit in $|1\rangle$
6. Measure all qubits in both cases
7. Compare the outputs

```
qc = QuantumCircuit(3,3)
```

```
qc1.cswap(0,1,2)
```

Deutsch-Jozsa Algorithm (Qiskit)

Step 1: Choose a small example (n = 2)

- Small enough to visualize
- Large enough to see superposition (4 inputs!)
- So: $x \in \{00, 01, 10, 11\}$

Step 2: Define two example oracles

We will test two cases:

- Case A — Constant oracle: $f(x) = 0$ for all x (you are also free to choose $f(x) = 1$ for all x)
- Case B — Balanced oracle: $f(x) = x_0 \oplus x_1$ 

x_0	x_1	$f(x)$
0	0	0
0	1	1
1	0	1
1	1	0

You are free to choose any other form of balanced $f(x)$. XOR is the simplest of operations that gives a balanced function.

Step 3: Algorithm steps

- Start in with $|0\rangle^{\otimes n}|1\rangle$
- Apply Hadamard gates $\rightarrow$ create superposition
- Apply the oracle once
- Apply Hadamard to the first register
- Measure only the first register

Function type	Measurement outcome
Constant	Always 00
Balanced	Never 00